

HEV

An abstract geometric composition featuring two triangles that meet at a central vertex. The upper triangle is solid black, and the lower triangle is a deep purple. To the left of this vertex, a small black rectangle contains the letters 'HEV' in a light gray, sans-serif font.



Architectures

HEV architectures

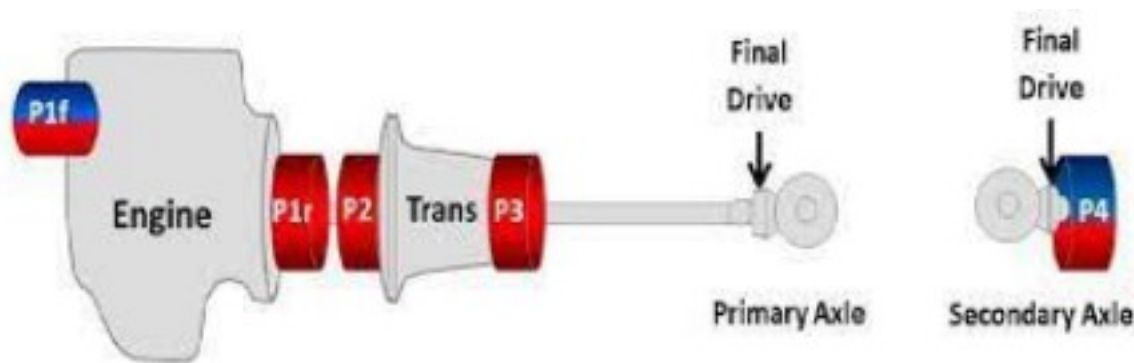
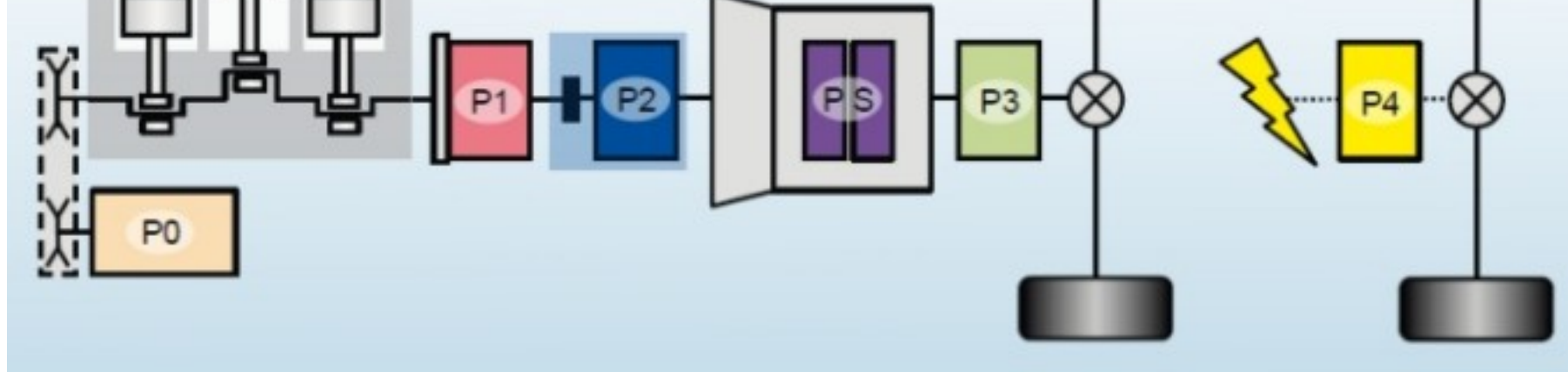
Hybrid vehicles

Today World had in large scale necessity to reduce CO2 emissions in big cities and for this all automotive group are changing the idea of business structuring them focal point on hybrid and electric solutions.

All automotive competitors accept this business challenge in different way and for this, today in the world there are a lot of different projects; the macro categories are: micro, mild, full and plug in hybrid vehicles and full electrics vehicles.

Note: we talk in this slides of hybrid where one of two motors is electric (not all hyb)





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Micro Hybrid

This type of vehicles are typically realized with an electric motor connected on belt slip and a secondary battery of 12V.

The electrical motor generate less than 1 cv and help thermal engine for few time in moments when thermal motor doesn't respect stechiometric value.

This vehicles can't generate a real driving of vehicle and help only thermal motor.

Fiat 500 Hybrid is an example of this type of vehicles but not the only



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BEV vehicles

A vehicle, in which all the propulsion is generated by an electric motor is commonly called BEV (Battery Electric

Vehicle)

The motor is connected to front axle through a single gear ratio: in these vehicles there's usually no need for a transmission (the motor is meant as a P4).

The new 500e is an example of BEV: it is equipped with a 70 kW/95 Cv motor, supplied by 400 V battery. The 500e motor can deliver up to 200 Nm torque; the expected range can reach 320 km



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BEV

BEV

The left schematic is taken from a Stellantis program and shows the main powertrain controllers used on

T

he BEV:

BPCM: HV battery pack controller estimates battery voltage, power limits and commands contactors, battery heating and cooling components

IDCM: this module is usually divided in two subsystems: OBCM and APM. OBCM is responsible for interfacing with external charger (EVSE) for HV battery; The APM controls the DC/DC converter for LV battery charging.

EVCU: is the main powertrain system controller; based on vehicle status and driver action, it generates inputs for the other modules and also computes the torque demands for the motor. This module also contains the standard vehicle functions as Cruise Control, Speed limiter, One Pedal Drive etc.

EDM: this module is responsible for translating the torque demand from EVCU into electrical inputs (phase currents and voltage) to drive the motor.

BEV

The electrical vehicles can use the motor to generate both positive and negative power. In this latter case, the motor is used as a generator to recharge the HV battery and thus to increase vehicle autonomy.

Find below some scenarios in which the motor can exploit some external energy source to regenerate the battery:

- **Regenerative braking (One Pedal Drive):** the electrical motor is used to brake the vehicle, converting the kinetic energy of the car into electrical energy for battery charging. When OPD is active the vehicle can be driven without using brake pedal: the motor generates a positive torque when pedal percentage is above the neutral point and a negative braking torque when the percentage is below the neutral position, thus the driver can perform almost any kind of maneuver by properly tuning the accelerator pedal position.
- **Cruise Control or Speed Limiter holding the vehicle on a negative grade:** on the BEV, CC and SL can hold the speed target even on high negative grades, thanks to the braking capability of the electrical motor. In this scenario the potential energy of the grade is converted and used to recharge the battery

REEV

The REEV (range extended electric vehicle) is a particular kind of hybrid, in which the Internal Combustion Engine (ICE) is only used for recharging the HV battery and thus to increase the vehicle expected range.

We can thus say that in the REEV, only series hybrid mode is available.

The typical REEV architecture consists of an electrical motor (a P4 as on standard BEV), meant for demanded torque actuation and an ICE, connected to a secondary electrical motor (a P0 or a P1). The latter works as a generator and uses the torque produced by the ICE to recharge the battery.

Below, you can find a REEV topology example.

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MHEV vehicles

Mild hybrid vehicles are vehicle that can have electrical motor in different positions, a type of configuration for MHEV vehicles can be P2.5 where electrical motor is in transmission and helps thermal engine during a gearshift.

This type of vehicles can drive in pure electric for few meters and for a vehicle speed less then 10 Km/h. For this boost electrical motors typically had a battery package of 48V

One example is Audi Q3 SB 1.5 TFSIe MHEV

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FHEV vehicles

Born in 1997 with Toyota is the real revolution in automotive world that have the first real big improvment to change the type of propulsions in vehicles.

During year this type of vehicles are realized for one or more electric vehicles and can have this electric motor in various positions.

FHEV can drive in only electric way for a some kilometers and different vehicle speeds (this limitation is decided from automotive customer). Full hybrid vehicles had a battery package more or less of 6Kw but the only possibility of charging is linked to kinetic energy.

In picture there is Toyota Prius the progenitor of FHEV

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PHEV vehicles

Plug in hybrid vehicles are the evolution of full hybrid vehicles because shared with them all architecture and technologies but have two biggest advantages: can be charged also with an external point and can have a battery package greater than FHEV vehicles because the dimension isn't limited to kinetic energy can be stored.

Now in market is the hybrid technology more publicized and present.

In this picture is showed Jeep Compass and Renegade 4xe that rappresent one of the possibilities for choise PHEV vehicles

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P1P4

EV propulsion
schematic where
thermal engine is off
and electrical engine is
the only that gives
movement to car

P1P4

PHEV P1P4 series
propulsion schematic
where thermal engine is
on but only for charge
battery package and
electrical engine is the
only that gives
movement to car

If available kinect
energy charge battery
package with P4 motor,
P1f motor charge
battery package using
thermal engine

P1P4

PHEV P1P4 parallel propulsion schematic where thermal engine is on and with electrical engine give movement to vehicle with maximum power available.

If available kinetic energy charge battery package with P1f motor and P4 motor

P1P2

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P1P2

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Network topology & ECUs

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